Name:		Centre/Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY **9744/03**

Paper 3 Long Structured and Free-Response Questions **22 September 2022**

Additional Materials: Answer Booklet **2 hours**

READ THESE INSTRUCTIONS FIRST:

Write your class, index number and name at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Sections A
Answer **all** questions in the spaces provided on the Question Paper.

Sections B
Answer any **one** free-response question in the Answer Booklet provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A / 50	
1	**26**
2	**16**
3	**8**
Section B / 25	
4 / 5	**25**
Total	**75**

This document consists of **15** printed pages and **1** blank page.

Section A: Long Structured Questions

Answer **all** questions.

Question 1

Fig 1.1 shows a male red deer, *Cervus elaphus*. Red deer are herbivores, browsing on low vegetation in forests and on waste land.

Fig 1.1

A history of wild red deer on the western European island of Ireland includes these facts:

- Red deer have lived in Ireland for at least 12 000 years.
- Originally, red deer could cross from the neighbouring island of Great Britain to Ireland over a land connection.
- A rise in sea levels at the end of the last Ice Age removed this land connection, separating the red deer on the two islands.

- In the 1800s, the number of red deer in Ireland decreased sharply after the main food crop for the human population failed for several consecutive years.
- In the 1900s, this decrease in the number of red deer continued as large areas of waste land were drained for agriculture.

- By 1960, red deer were nearly extinct in Ireland, restricted to one population, **A**, of 60 individuals.
- Since then, protection has allowed population **A** to increase to over 600 red deer.
- Several new red deer populations, **B**, **C** and **D**, have also been established in different parts of Ireland from individuals brought from Great Britain.

(a) **(i)** Use the information given to identify two causes of extinction that may have threatened the survival of red deer in Ireland. [2]

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(ii) Describe how the level of molecular similarity between the red deer in population **A** and population **B** can be investigated. [3]

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(iii) Explain why some red deer in population **A** show unique molecular features that are not found in any of the red deer in populations **B**, **C** and **D**. [3]

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The mitochondrial cytochrome *b* (cyt-b) gene sequences (1140 bp) of one subspecies of the European red deer (*Cervus elaphus* in Europe), three subspecies of the wapiti (*C. elaphus* in Asia and North America), and six subspecies of the sika deer (*C. nippon* in Japan) were compared. The wapiti was shown to be more closely related to the sika deer than to the European red deer. This is in conflict with traditional morphological results, which suggest a close sister group relationship between the wapiti and the European red deer.

(b) Suggest why the mitochondrial cytochrome *b* (cyt-b) gene is used to compare the phylogenetic relationships between the three subspecies of deer. [2]

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The domestic dog, *Canis familiaris*, is found worldwide. It is able to breed with all other members of the genus to form fertile hybrids.

Table 1 shows whether members of different species of the genus *Canis* are able to breed with each other.

Table 1

key: ✓ = able to interbreed ✗ = unable to interbreed **?** = interbreeding unknown

	dingo	**grey wolf**	**golden jackal**	**side-striped jackal**	**black-backed jackal**	**domestic dog**
dingo	✓	**?**	**?**	**?**	**?**	✓
grey wolf	**?**	✓	**?**	**?**	**?**	✓
golden jackal	**?**	**?**	✓	✗	✗	✓
side-striped jackal	**?**	**?**	✗	✓	✗	✓
black-backed jackal	**?**	**?**	✗	✗	✓	✓
domestic dog	✓	✓	✓	✓	✓	✓

(c) Suggest why the members of the genus *Canis* could be described as one species. [2]

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Using molecular techniques and comparing the genomic sequences, a phylogenetic tree of the members of the genus *Canis* was constructed, as shown in Fig 1.2.

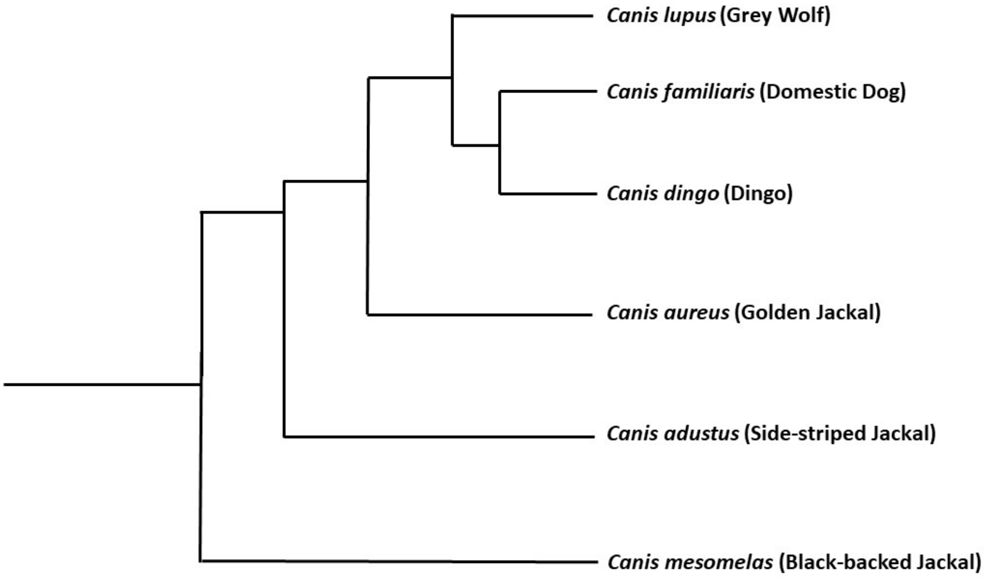

Fig 1.2

(d) **(i)** Explain why using molecular techniques is better than using the information of interbreeding to determine the evolutionary relatedness between the members of the genus *Canis*. [2]

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(ii) Explain which of the two remaining species of jackals is more closely related to the golden jackal. [1]

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The distribution of some of the species belonging to the genus *Canis* is shown in Fig 1.3.

The dingo and the grey wolf species have distinct ranges in Australia and Europe respectively.

Fig 1.3

(e) Together with information from Fig 1.2 and Fig 1.3, explain how dingo could have evolved into a distinct species. [5]

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The Hawaiian Islands are some of the most isolated islands in the world. They are made up of islands that are formed at different times. The first birds to have flown to these islands probably arrived millions of years ago from East Asia.

Fig 1.4 and Fig 1.5 show the fossils of two extinct species of Hawaiian waterfowl found on two different islands. The giant Hawaiian goose was a flightless bird whereas the nene could fly. Until recently, the evolutionary relationships among Hawaiian waterfowl are known only from bone structures. Fig 1.4 shows the skulls and mandibles while Fig 1.5 shows the wing and leg bones of the giant Hawaiian goose and nene.

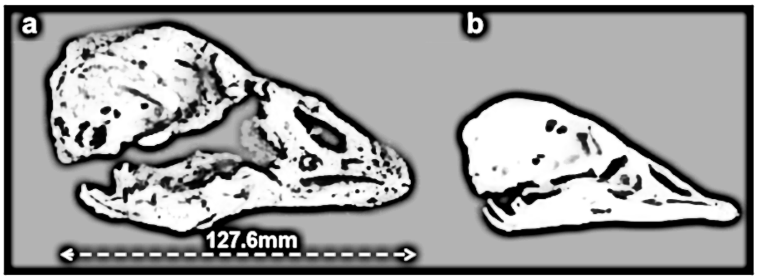

Fig 1.4 : Skulls and mandibles of (a) Hawaiian goose and (b) nene

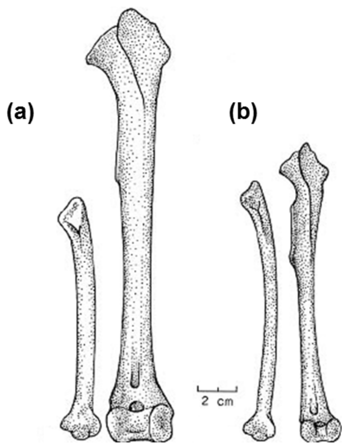

Fig 1.5 : Wing (left) and leg bones (right) of (a) giant Hawaii goose and (b) nene

(f) Based on the fossils, state two species concepts that can be used to determine whether the Hawaiian goose and nene belong to the same species. [2]

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(g) With reference to Fig 1.4 and Fig 1.5, discuss whether these fossils can be used to support Darwin's theory of evolution. [4]

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Total: [26]

Question 2

A global strategy to tackle malaria involves the rapid diagnostic testing (RDT) of individuals who may have malaria. This involves testing human blood samples for the presence of proteins specific to *Plasmodium*. RDT test sticks make use of monoclonal antibodies (mAbs).

mAbs are antibodies that are all identical to each other. mAbs are produced *in vitro* by fusing a plasma cell with a cancer cell to produce a hybridoma, which divides repeatedly to form many genetically identical cells that all produce the same antibodies.

Table 2 contains information about two RDT test sticks.

Table 2

Test stick	*Plasmodium* protein tested for	Species of *Plasmodium* that produce the protein
1	pLDH (parasite lactate dehydrogenase)	*P. vivax* *P. falciparum* *P. ovale* *P. malariae*
2	HRP-2 (histidine-rich protein 2)	*P. falciparum* only

Some details of the design of these RDT test sticks are shown in Fig 2.1.

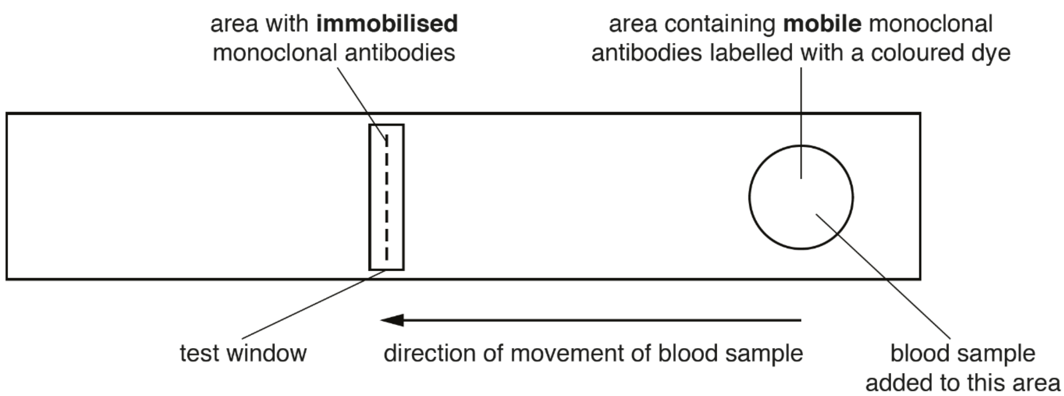

Fig 2.1

The **immobilised** monoclonal antibodies in the test window are not visible.

If the blood sample contains a *Plasmodium* protein that can be detected by the RDT test stick,

- the **mobile** monoclonal antibodies bind to one part of the protein,
- the **immobilised** monoclonal antibodies bind to another part of the protein,
- a coloured line in the test window indicates a positive result for the protein.

(a) With reference to Table 2 and Fig 2.1, [2]

(i) explain why test stick 1 and test stick 2 will contain different mobile monoclonal antibodies,

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(ii) explain what can be diagnosed for this person from a positive result for test stick 1 and a negative result for test stick 2. [2]

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(b) Outline the process during B-cell development that allow our immune system to produce antibodies that recognise a range of *Plasmodium* proteins. [4]

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Prokaryotic cells can potentially be used for antibody production. Fig 2.2 shows a possible experimental procedure for antibody production in bacteria cells.

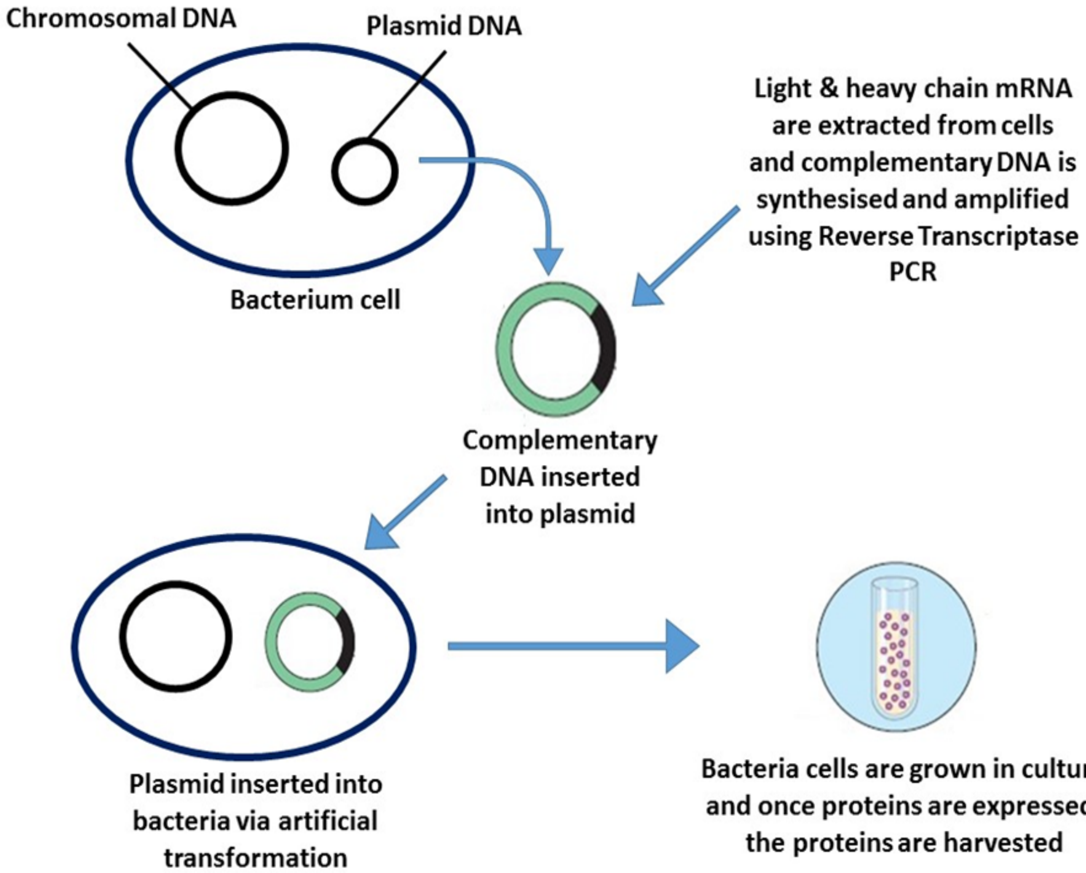

Fig 2.2

To obtain DNA sequences of the light and heavy chain genes, light and heavy chain mRNA are extracted from plasma cells and then DNA containing the gene sequences are produced using reverse transcription Polymerase Chain Reaction (PCR).

The procedure for reverse transcription PCR is as follows:

- The extracted mRNA is mixed with deoxyribonucleotides, reverse transcriptase and incubated at 65°C for 5 minutes.
- RNase is then added to the reaction mixture and incubated at 42°C for 50 minutes, then 70°C for 15 minutes.
- Primers, *Taq* polymerase and deoxyribonucleotides are then added. The first cycle of PCR is carried out at 72°C, before the normal three-stage PCR cycle is carried out for the next 40 cycles at temperature 95°C, 60°C and 72°C.

(c) For the reverse transcription PCR procedure, explain why

(i) extracted mRNA is mixed with deoxyribonucleotides and reverse transcriptase; [1]

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(ii) RNase is added to the reaction mixture and eventually incubated at 70°C for 15 minutes; [2]

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(iii) the first cycle of PCR is carried out at 72°C instead of the usual 95°C. [2]

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(d) Production of fully functional antibodies, which originate from eukaryotic cells, in prokaryotic cells is expected to be unsuccessful. [3]

Explain why production of fully functional antibodies in prokaryotic cells is expected to be unsuccessful.

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Total: [16]

Question 3

Plant biodiversity varies throughout the world and is dependent on many factors, particularly climate.

Fig 3 shows the relationship between the number of plant genera and the mean annual rainfall in seven countries.

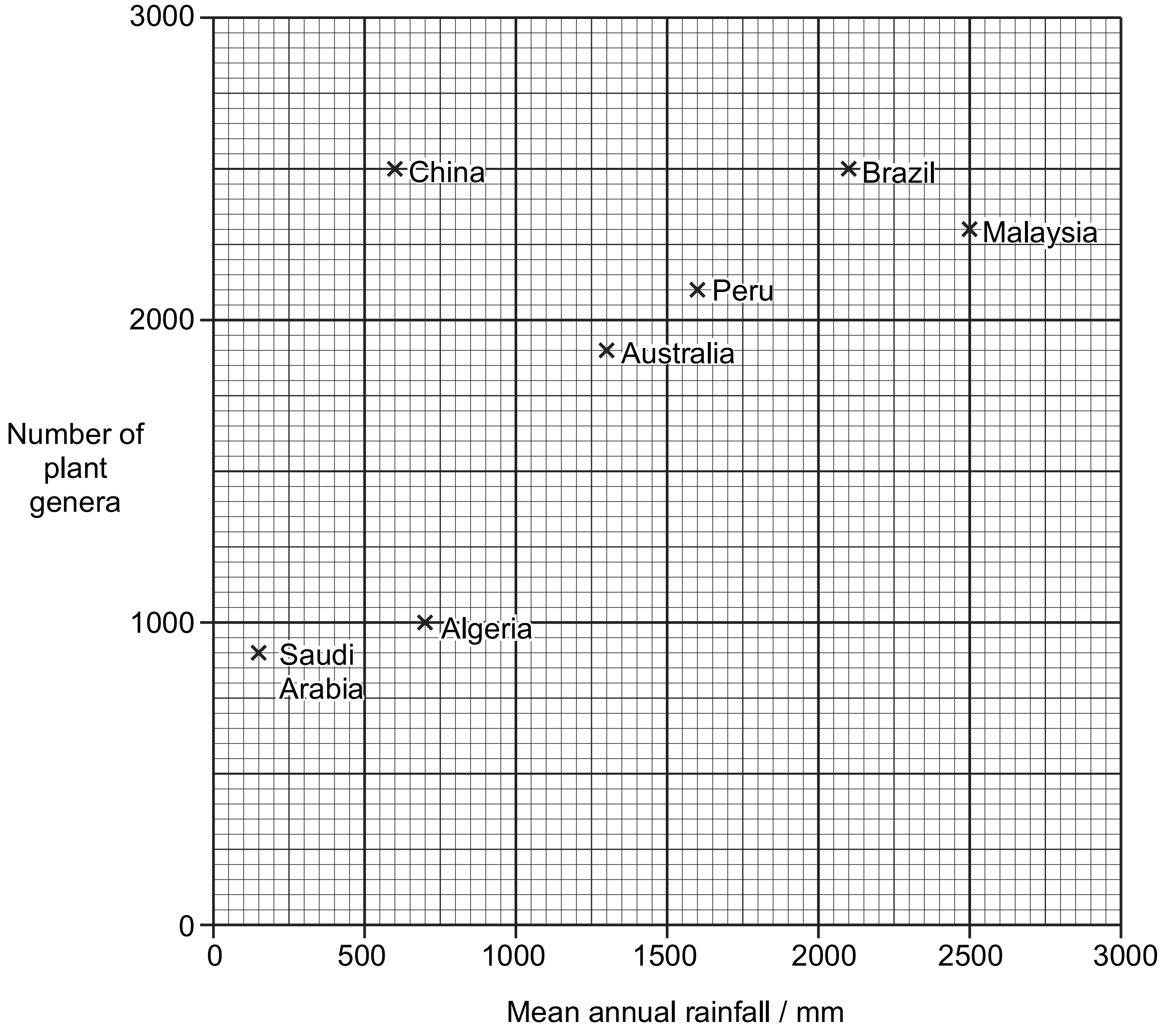

Fig 3

(a) **(i)** Describe the relationship between the number of plant genera and the mean annual rainfall in these seven countries. [2]

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(ii) Discuss how changes in rainfall can affect plant biodiversity. [3]

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(b) Suggest the benefits to humans of conserving plant species. [3]

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Total: [8]

Section B: Free-Response Questions

Answer **one** question.

Write your answers in the Answer Booklet provided.
Your answers should be illustrated by large, clearly labelled diagrams, wherever appropriate.

Your answer must be in continuous prose, where appropriate.
A **NIL RETURN** is required.

4 **a** In eukaryotes, a gene mutation can give rise to a new allele. Outline how the frequency of a gene mutation can increase in a population. Consider both prokaryotes and eukaryotes. [10]

b Describe how the structures of nuclei acids are adapted for protein synthesis. [15]

5 **a** Using named examples, outline the role of ATP in eukaryotes. [10]

b Eukaryotic cells contain various organelles that perform different specific functions. These organelles comprise membranes that have different compositions that are adapted to their specific roles. [15]

Describe the different structures of various membranes using named examples and explain how they are adapted to the role they perform in a eukaryotic cell.